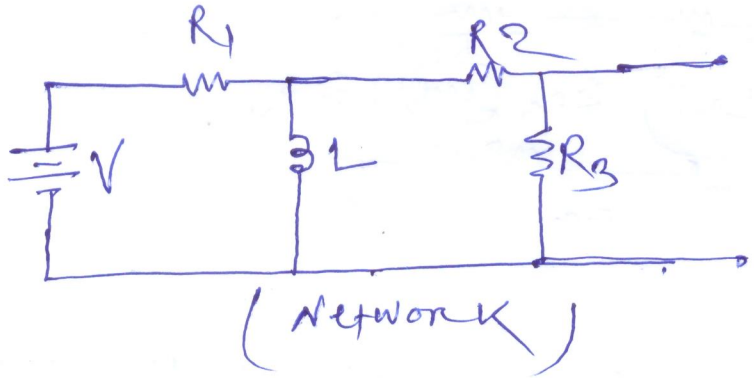
Lecture-01

Circuit :-

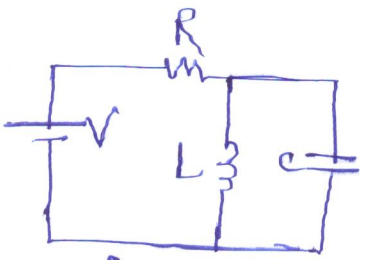
→ Any combination of electrical and electronics element are called as circuit.
→ It is always a close one.

Network :-

→ Any combination of electrical and electronics element for a specific purpose and current can be flow through it.
→ It may be open or close.
→ e.g.,



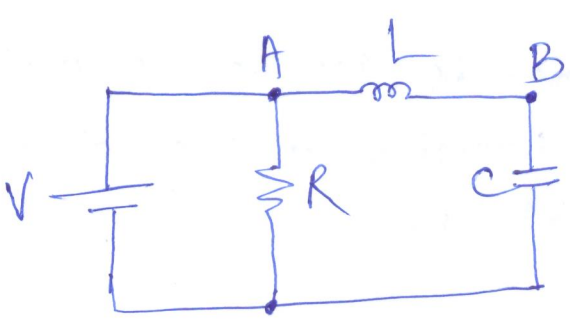
(Network)



(Circuit)

Node :-

→ A node is a point where 2 or more n/w branches are joined together.



Here 'A', 'B' are nodes

'C' is known as reference node b/c most of the branches are connected to this node.

Branch :-

→ It is the part of network which lies between two node point.

Mesh:-

→ It is a loop in the network which does not contain any other loop.

Loop:-

It is a closed path in a network in which no. element and no node point is encounter or travelled more than one.

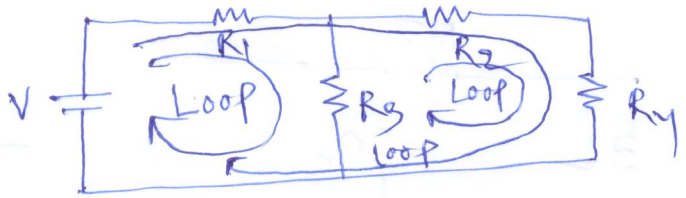


Fig.

Potential:-

The electric potential at a point is the amount of work done by taking a unit positive charge from infinite to that point against the field. ie

$$V = \frac{W}{q} \Rightarrow W = Vq$$

Ex→ The potential of point is 1V ie 1 Joule if work is done by bringing a 1 c of positive charge from ∞ to that point.

Potential Difference:-

The potential difference is exist between 2 points It is defined as the amount of work done by moving a u.p.f from one point to another point.

EMF:-

This is the force require to overcome the resistance of conductor which is generally produced a generator or battery.